

SDI Training for ICT Professionals

QGIS Map Making

Styling, Labelling, and Print-Ready Maps

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Day 4 | 30 Minutes Lecture + Extended Practical | crd.resilienceacademy.ac.tz

Styling Layers: Categorised and Graduated Symbols

Categorised Symbols

Use when your data has **distinct categories** (e.g., facility type: school, hospital, market).

- 1 Right-click layer → **Properties**
- 2 Go to **Symbology** tab
- 3 Change “Single Symbol” to **Categorized**
- 4 Choose the **Column** (e.g., `facility_type`)
- 5 Click **Classify** — each category gets a unique colour
- 6 Click **OK**

Graduated Symbols

Use when your data has **numeric values** (e.g., number of students, flood depth).

- 1 Right-click layer → **Properties**
- 2 Go to **Symbology** tab
- 3 Change to **Graduated**
- 4 Choose the **Column** (e.g., `num_students`)
- 5 Set number of **Classes** (e.g., 5)
- 6 Choose a **Color ramp** (e.g., green to red)
- 7 Click **Classify**, then **OK**

The Attribute Table: View, Filter, Select

Opening the Attribute Table

- Right-click on a WFS layer → **Open Attribute Table**
- You see a spreadsheet of all features and their properties
- Each **row** = one feature (one school, one point)
- Each **column** = one attribute (name, type, location)

Selecting Features

- Click a row to **select** it — it highlights on

Filtering with Expressions

Example: find all education facilities in Ilemela district.

- 1 Open Attribute Table
- 2 Click the **Select by Expression** icon (yellow square with ε)
- 3 Type: "district" = 'Ilemela'
- 4 Click **Select Features**
- 5 Selected features turn **yellow** on the map

Remember

Adding Labels to Your Map

How to Add Labels

- 1 Right-click layer → **Properties**
- 2 Go to the **Labels** tab
- 3 Change “No Labels” to **Single Labels**
- 4 Choose the **Value** column:
 - For wards: select `ward_name`
 - For facilities: select `facility_name`
- 5 Click **OK** — names appear on the map

Improving Label Appearance

In the Labels properties, you can adjust:

- **Font** — size, colour, bold/italic
- **Buffer** — white outline around text for readability (enable under *Buffer* sub-tab)
- **Placement** — above, below, or centred on the feature
- **Rendering** — show labels only at certain zoom levels

Best Practice

Print Layout: Creating a Professional Map

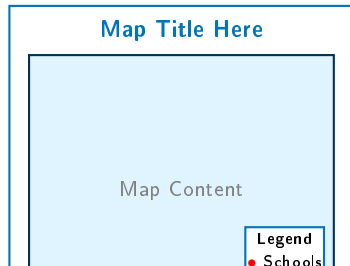
Opening the Print Layout

- 1 Go to **Project** → **New Print Layout**
- 2 Give it a name (e.g., “Mwanza Education Map”)
- 3 A blank page opens — this is your map canvas for printing

Adding Map Elements

Use the left toolbar to add items:

- **Add Map** — draw a rectangle; your map appears inside it
- **Add Title** — click *Add Label*, type your title, make it large and bold
- **Add Legend** — shows what each



Exporting Your Map as PDF or PNG

Export Options

In the Print Layout, go to the **Layout** menu:

- **Export as PDF**
 - Best for **printing** and **formal reports**
 - Vector quality — sharp at any zoom level
 - Recommended resolution: 300 DPI
- **Export as Image (PNG)**
 - Best for **presentations** and **web use**
 - Fixed pixel size
 - Recommended resolution: 150–300 DPI

Step-by-Step Export

- 1 Ensure your layout looks correct
- 2 Go to **Layout** → **Export as PDF...**
- 3 Choose a save location and file name
- 4 Set resolution to **300 DPI**
- 5 Click **Save**
- 6 Open the PDF to verify — it should look crisp and professional

Save Your Project

Before exporting, save your QGIS project
(Ctrl+S). This saves all layer connections

Session Summary

Map Making Workflow in QGIS

- 1 **Load data** — connect to CRD via WMS/WFS (Session 1)
- 2 **Style layers** — categorised or graduated symbols to show meaning
- 3 **Explore data** — use the attribute table to filter and select features
- 4 **Add labels** — show ward names, facility names on the map
- 5 **Create print layout** — add title, legend, scale bar, north arrow
- 6 **Export** — save as PDF for reports or PNG for presentations

Practical Exercise (Next)

You will now create your own professional map:

Key Shortcuts

- **Ctrl+S** — Save project
- **F5** — Refresh map